

Paired Uniform Cubic B-Splines are Strong Approximations to Represent Skeleton Activity

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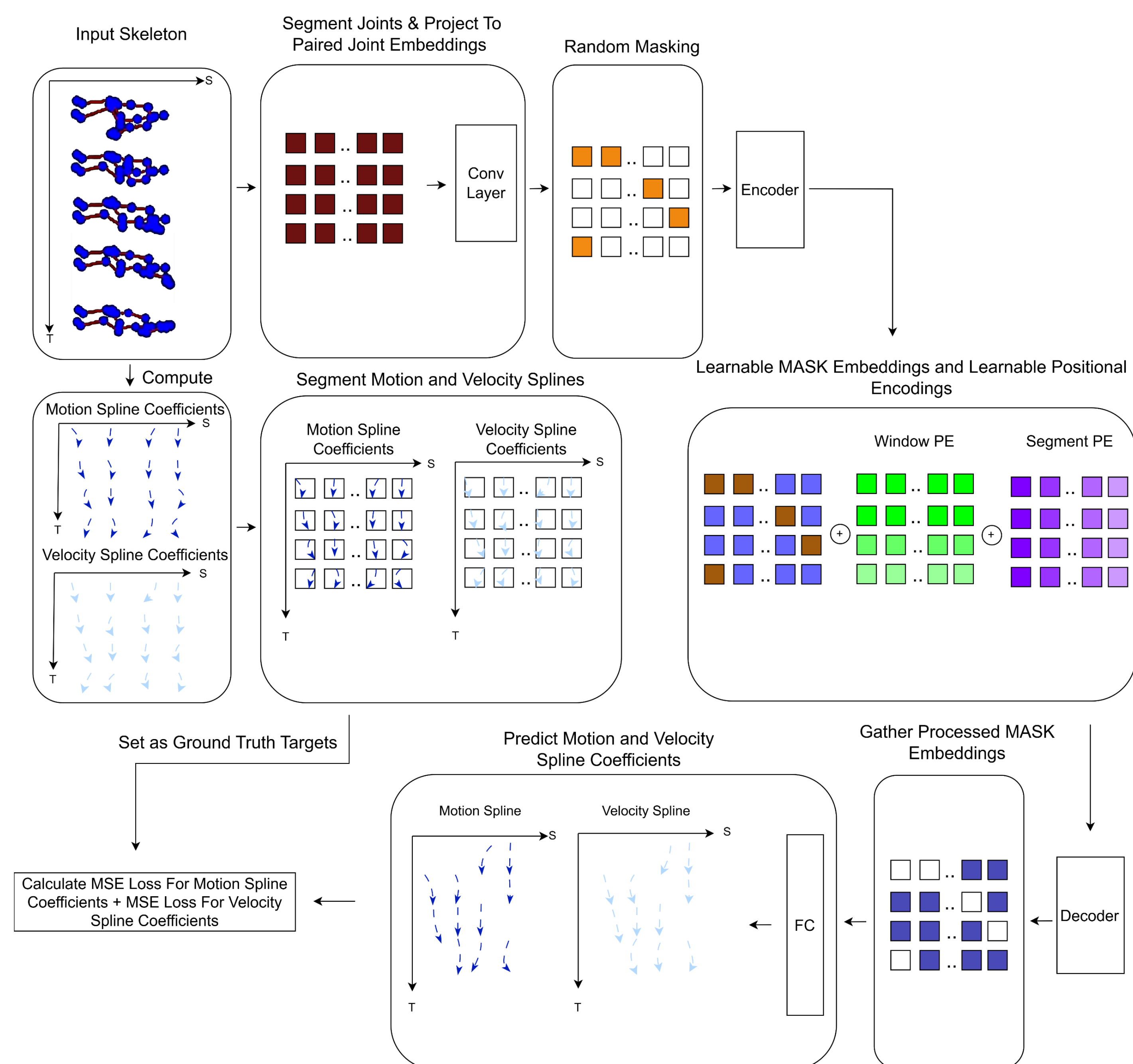
1) Summary:

We propose our **Masked Paired Spline Coefficient Predictor (MPSCP)** to learn the **smoothened motion** of paired skeleton joints using the **coefficients** of the **B-Spline segments** to **represent** the **spatial pattern** and the **velocity** of the motion.

2) Motivation:

- ❑ In a **live environment**, **no labels** are **available** for autonomous agents to learn using **supervised learning**.
- ❑ Methods utilising **RGB and Depth Imagery** have produced **state-of-the-art results (SOTA)** but **compromise** the **privacy** of the subjects recorded.
- ❑ Most **unsupervised** and **self-supervised** SOTA models (especially transformer-based models) require a **substantial** number of **resources** and **time**.
- ❑ We took **inspiration** from **3D animation software**, where **spline-based techniques** have been used to **generate smooth, natural motion** by **approximating curves** through **keyframes** on a **bone-rigged character model**.

3) Our Pre-training Pipeline



4) Linear Evaluation Protocol (Accuracy):

Model	Input	NTU 60		NTU 120		PKU-MMD	
		XSub	XView	XSub	XSet	Phase I	Phase II
LongT GAN	J only	39.1	48.1	-	-	67.7	26.0
MS^2L	J only	52.6	-	-	-	64.9	27.6
P&C	J only	50.7	76.3	42.7	41.7	59.9	25.5
ISC	J only	76.3	85.2	67.1	67.9	80.9	36.0
3s-Skeleton CLR	J+M+B	75.0	79.8	60.7	62.6	85.3	-
3s-CrosCLR	J+M+B	77.8	83.4	67.9	66.7	84.9	21.2
3s-AimCLR	J+M+B	78.9	83.8	68.2	68.8	87.4	39.5
CPM	J only	78.7	84.9	68.7	69.6	88.8	48.3
3s-RVTCLR+	J+M+B	79.7	84.6	68.0	68.9	-	-
SkeletonMAE	J only	74.8	77.7	72.5	73.5	82.8	36.1
MAMP	J only	84.9	79.1	78.6	79.1	92.2	53.8
Ours	PJ only	84.6	91.3	70.5	74.0	89.0	54.7

5) Efficiency Comparison:

Model	Accuracy (%)	Pre-train Duration (hr)	Inference Duration (hr)	Parameters (M)	Embeddings per Activity	Pre-train VRAM Used (GB)	LEP VRAM Used (GB)
Skeleton MAE	74.8	13.5	3.23	8.7890	750	56.0	18.8
MAMP	84.9	20.5	3.37	8.7890	750	79.6	18.8
Ours	84.6	10.6	2.88	8.0784	192	6.5	5.7

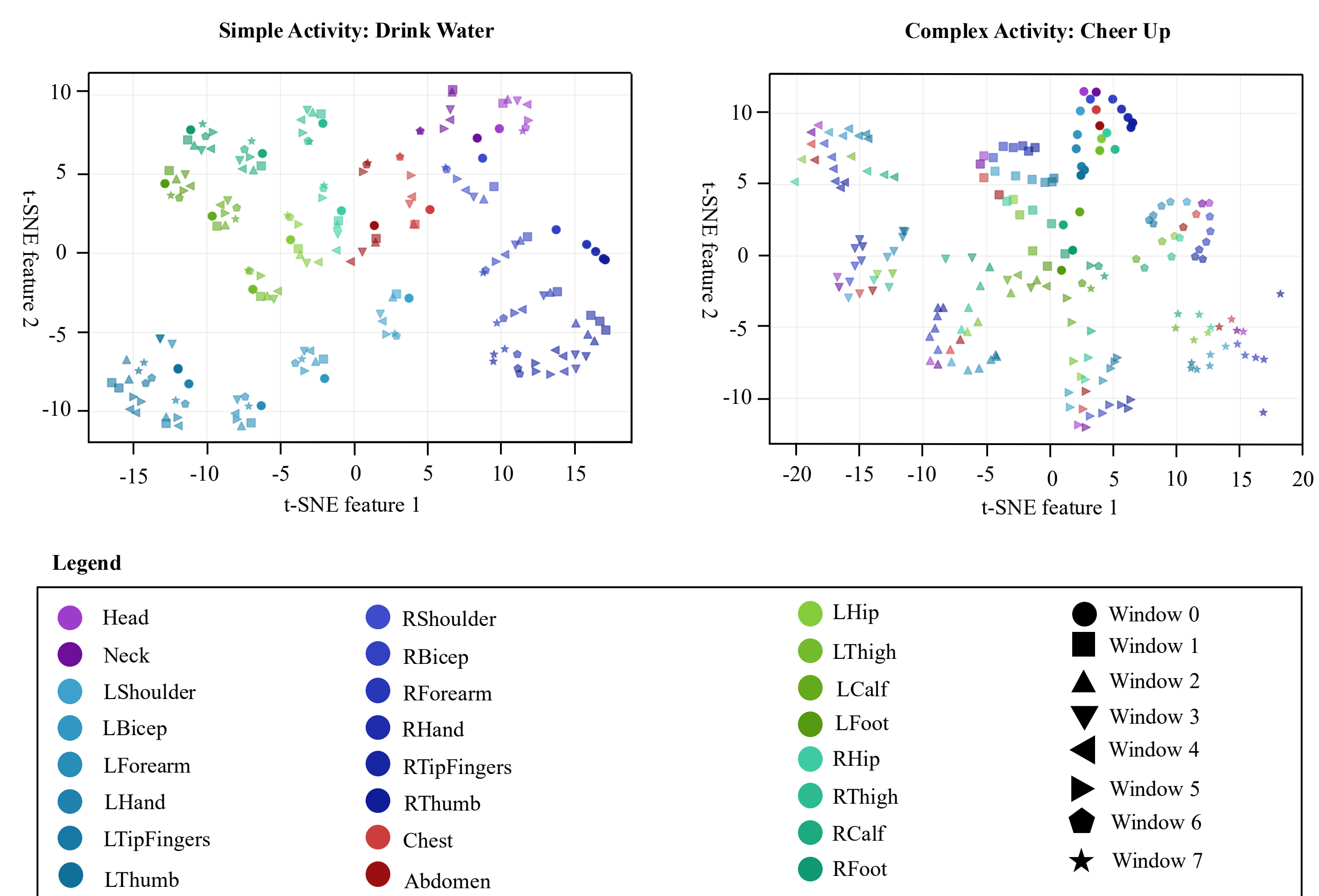
- ✓ Our model has obtained **approximately** the **same results** as the **SOTA** on the NTU 60 Cross Subject dataset.
- ✓ Our model used **less VRAM** and **less computing time** during **pre-training** and **inference**.

6) Is the Positional Encoding at the Encoder Necessary?

Combination	No PE	Encoder PE Only	Decoder PE Only	Both PE
Accuracy	47.4	45.9	54.7	54.1

- ✓ The **representation learnt** using **only positional encoding** at the **decoder** achieved the best results.
- ✓ **Pairing joints** and using **windows** of the paired joints to **represent** the motion through embeddings **removes** the **need** for **positional encoding** at the **encoder**.

7) Visualisation of Embeddings in Latent Space:



- **Simple** activities tend to exhibit **motion patterns over time**.
- **Complex** activities seem to show **pose-like patterns** in each **window of time**.

8) Key Takeaways:

- ✓ Using **coefficients** of the **motion** and **velocity splines** reduces the **computing resources needed** and **shortens pre-training** and **inference duration** while maintaining **performance comparable** to **state-of-the-art models**.
- ✓ **One positional encoding** at the **decoder** is enough.

